ML-Powered Predictive Inventory Management using FaceBook’s PROPHET – Proof of Concept

Bird's Eye View of POC

Visual walkthrough of the prototype implementation

Database schema snapshot
Entity-relationship diagram highlighting SKU, timestamp, and quantity tables.

Inventory dataset view
Tabular preview of raw historical records used for model training.

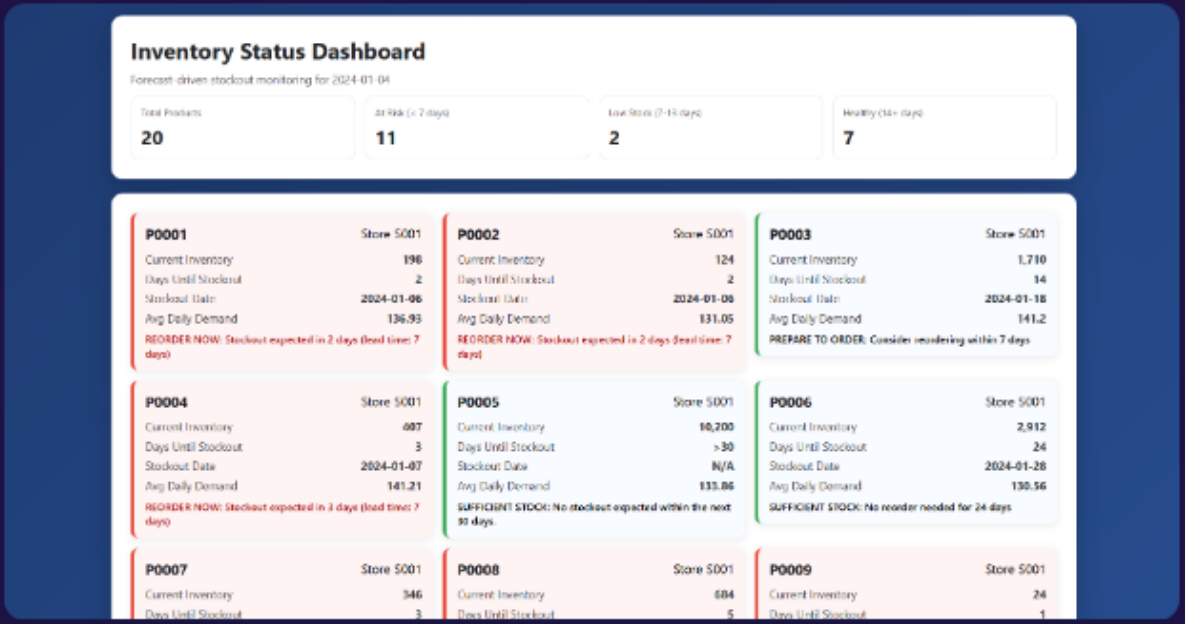
Data pipeline script overview
Annotated Python code that transforms raw rows into Prophet time-series objects.

Forecast execution console
Command-line output showing model fitting progress and forecast generation timestamps.

POC Screenshots

Warehouse Level

- **1:** Each SKU/product is visible for the particular warehouse along with its status
- **2:** Each status shows a products stockout date ahead of time by a time frame of 30 days
- **3:** A filter function and a notification function can be added when a product reaches at risk status
- **4:** Within the time-frame window (of 30 days) the Prophet model is run and stockout is predicted on a 7 day basis



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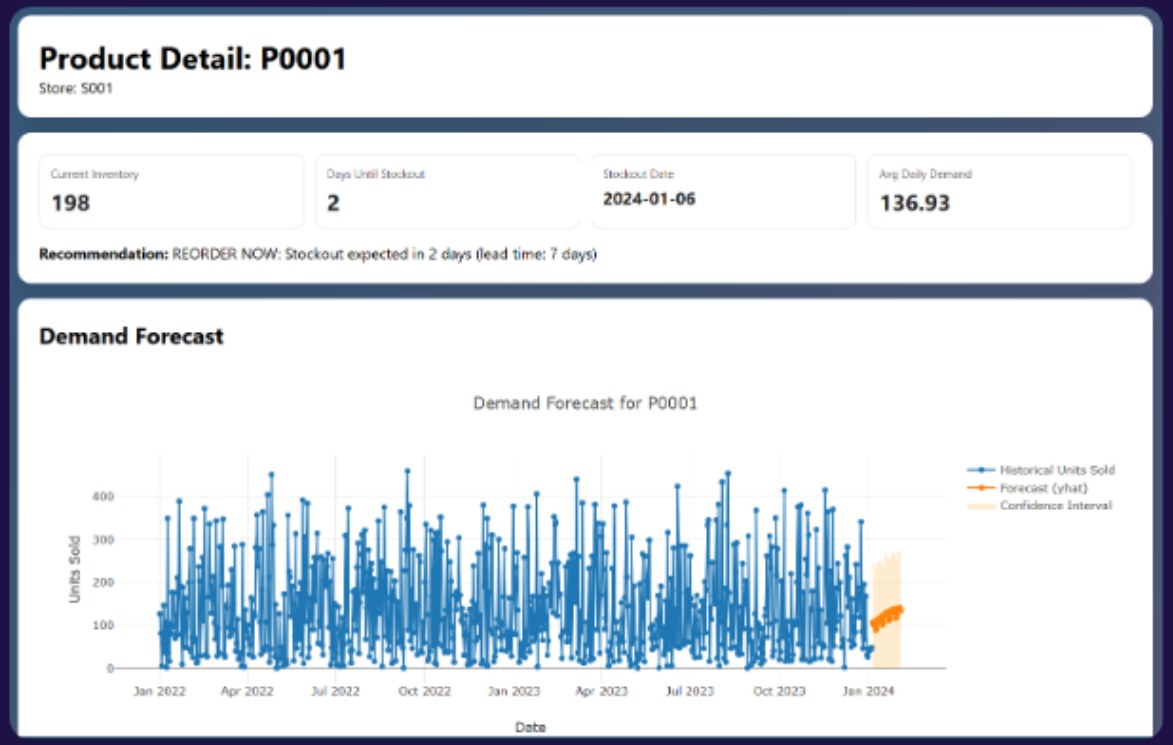
POC Screenshots

Product Level



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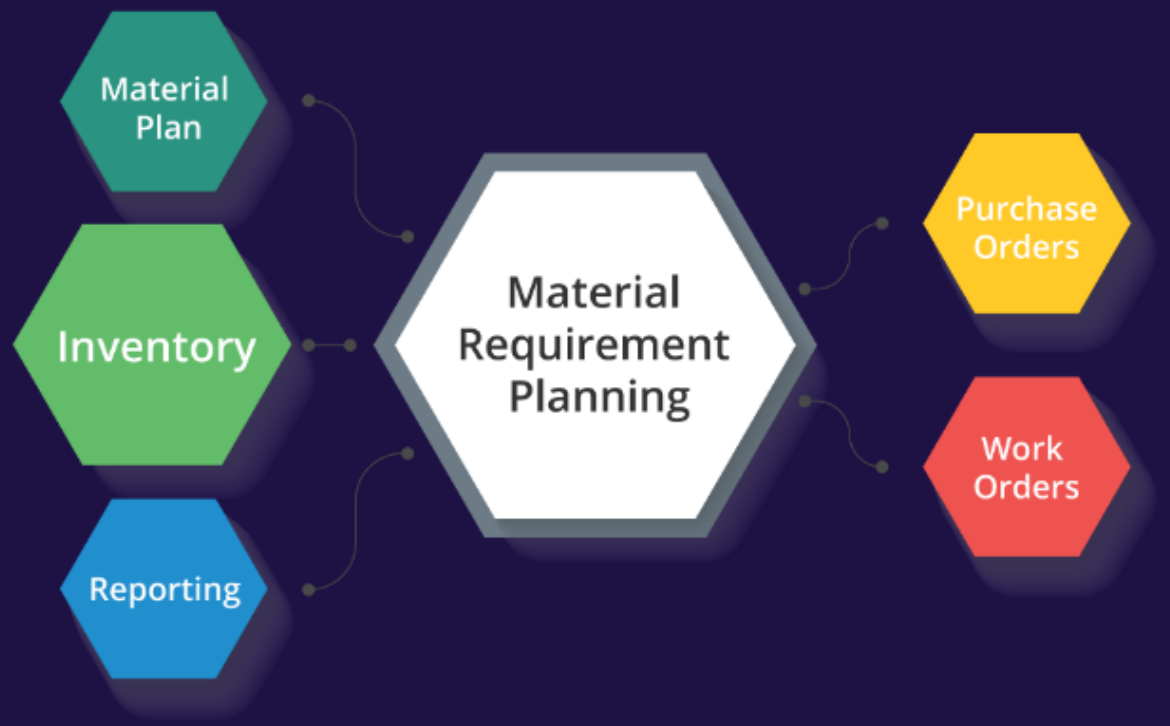
- **1:** Each product can be viewed at at individual level
- **2:** Status/Sales history shown in blue; Forecast for next 30 days shown in yellow
- **3:** Predicted stockout date is shown on top
- **4:** NOTE: on page refresh, live data is accounted for i.e if stockout date for SKU is approaching and live data stating restock has occurred, the prediction will RE-COMPUTE in REAL TIME

The Inventory Planning Challenge

Core pain points of current inventory planning

- **Reactive inventory systems:** Replenishment is triggered only after inventory drops below a threshold, leading to delayed responses.
- **Manual/average-based decisions:** Planners rely on static averages rather than dynamic demand signals, increasing forecast error.
- **Frequent stockouts:** Out-of-stock events occur regularly, harming service levels and customer satisfaction.
- **Overstocking costs:** Excess inventory ties up capital, inflates holding costs, and risks obsolescence.
- **Inefficient warehouse utilization:** Space is either under-used or congested, reducing picking efficiency and increasing labor.
- **Limited SKU-level demand visibility:** Granular demand patterns are hidden, preventing precise allocation per product.
- **Need for proactive predictive layer:** A forward-looking model is required to anticipate demand before stock levels change.



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Potential Future Upgrades

A few things we can add

- **Performance:** Prevent re-compute on page reload everytime, have indexes in the DB for fast lookups and caches on local machines
- **Visualization (Power BI):** We can add the forecasting graph data into a reporting tool such as Power BI to gain better insights, make better business decisions etc